

CBAM certificate estimator

Algorithms, equations and rounding, prepared for external review

VERSION 1.1 · 31 JULY 2026 · ENGINE SNAPSHOT: RULE PACK 2026-07-29

Scope of this document. It states exactly what the estimator computes, which published instrument each value comes from, and where the calculation deliberately refuses to produce a number. Every figure below was traced from the running engine rather than reconstructed from documentation.

Status of the tool. Screening grade decision support. It is not a filing, it is not validated by the EU CBAM Registry, and it is not a statement of monetary liability. It runs entirely in the visitor's browser from Commission default values; no input is transmitted.

Changes in version 1.1. Section 2.2 is new. It states which calculation paths the browser estimator does *not* take, and why, with the size of the Column A / Column B divergence measured across the shipped rule pack. Added in response to external review, which correctly identified Column A as the boundary of what this tool does. No figure, equation or rounding rule has changed.

1. Computation chain

```

CN code + origin + production route + net mass + import date + scope
|
+--> default intensity      I = baseIntensity x (1 + markupPct/100)
|   direct emissions       E = I x mass
|   indirect emissions     G = I_ind x mass          (opt-in; 3 sectors only)
|
+--> SEFA per tonne        S = CBAM_y x CSCF_y x SUM(BM_i x q_i)
|   free allocation        F = S x mass
|
+--> chargeable            N = max(0, E - F) + G
+--> certificates          C = N                      (1 certificate = 1 tCO2e)
+--> cost                  EUR = C x price_quarter

```

2. Governing equations

SEFA, the specific embedded free allocation, per Implementing Regulation (EU) 2025/2620:

```

Eq 2 (actual data)   SFAProc = CBAM_y x CSCF_y x BM_A(process)
Eq 4 (complex good)  SEFA    = SFAProc + SUM(mass_i x SEFA_i)
Eq 6 (default path)  SEFA    = CBAM_y x CSCF_y x BM_B(good)

```

This calculator runs **Equation 6 exclusively**. It uses Commission default values, so the emissions figure is all inclusive and the deduction must be the full product Column B benchmark.

2.1 Column A against Column B

This is the most consequential decision in the engine. It is driven by the *scope of the emissions figure being deducted from*, not by whether the data was verified.

SCOPE	COLUMN	REASON
full_product	B	The figure already embeds the good's precursors, so the deduction must be the full product benchmark.
process_only	A + Eq 4	The figure covers this installation's process alone and precursors are carried separately, so the deduction is the process benchmark plus each precursor's own Column B value.

Why it matters. Deducting a process only benchmark from an all inclusive emissions figure leaves precursor emissions in the chargeable total with no matching free allocation, over charging the importer several fold. For a cement grinder, whose Column A is zero, it removes the deduction entirely.

The engine also refuses the inverse error outright. A `full_product` scope submitted with precursors raises `COLUMN_B_ALREADY_COVERS_PRECURSORS`, because Column B already covers them and adding precursors on top would double count the deduction and under charge.

2.2 What this estimator does not do, and why

The browser estimator takes the **defaults path only**. Its input construction is fixed, at `estimator/estimate-from-pack.ts`:

```
scope:      'full_product'    // -> Column B, on every line
precursors: []                // -> Eq 4 never runs
tier:       'default+markup'
emissionsType: 'direct'
```

Equation 2 and Equation 4, and with them the whole Column A path, are **fully implemented** in `cbam/sefa.ts`. They are the same audited code the GeoCBAM ledger runs. They are simply not reachable from this tool, by design.

The reason is what Column A means. A Column A deduction is only correct against a *process only* emissions figure with its precursors declared separately: an operator's own verified data plus a precursor list. A public screening tool has no way to obtain either. Offering the choice would invite users to select Column A against a Commission default, which is all inclusive, and that is the expensive error described in 2.1.

The size of the divergence, measured on the shipped pack (2026-07-29), so the consequence of choosing wrongly is not left as an assertion:

MEASURE	VALUE
Column A benchmark rows in the pack	661
Goods publishing <i>both</i> a Column A and a Column B value	141
Of those, goods where A does not equal B	102 (72%)
Column A values that are exactly zero	9

The zero rows are the clearest case. Three of them are cement grinding goods, where the Column B value is substantial:

CN	ROUTE	COLUMN A	COLUMN B	GOOD
25232100	any	0	0.859	White Portland cement
25232900	any	0	0.666	Other Portland cement
25239000	(A)	0	0.666	Other hydraulic cements
25239000	(B)	0	0.847	Other hydraulic cements
28080000	any	0.151	0.582	Nitric acid

On a 100 tonne line of CN 25232900, taking Column A against a default emissions figure would value the free allocation at **zero** instead of 0.666 tCO₂e per tonne. At the 2026 CBAM factor and the 2026-Q1 certificate price that is a deduction of roughly 64.9 tCO₂e, about €4,890, silently withdrawn from the importer's favour.

Where Column A does belong. The operator data path, in the workspace product, where a declarant supplies a verified process only figure together with the precursor quantities the good actually consumed. The engine is ready for it. Exposing it in a public screening tool would require taking precursor inputs from users who mostly do not have them, and every wrong entry moves the bill in the importer's disfavour without warning. That is a product decision, not an arithmetic one, and it is deliberately not taken here.

3. Worked example, fully traced

CN 25231000 (cement clinker), origin Algeria, production route (A), 100 tonnes, import date 2026-03-15, scope direct and indirect.

default factor	baseIntensity 1.24	markupPct 10	type direct
I = 1.24 x 1.10		= 1.364	tCO2e/t
E = 1.364 x 100		= 136.4	tCO2e
indirect factor	baseIntensity 0.06	markupPct 10	
G = 0.06 x 1.10 x 100		= 6.6	tCO2e
benchmark, Column B, route (A), scope 25231000		= 0.666	tCO2e/t
IR (EU) 2025/2620 Annex, Column B route (A)			
CBAM factor 2026		= 0.975	
Dir 2003/87/EC Art 10a(1a), free allocation retained in 2026			
CSCF 2026		= null, status PENDING	
DR (EU) 2019/331 Art 14(6), no CSCF published for 2026			
S = 0.975 x 1 x 0.666		= 0.64935	tCO2e/t [CSCF assumed 1]
F = 0.64935 x 100		= 64.935	tCO2e
N = max(0, 136.4 - 64.935) + 6.6		= 78.065	tCO2e
C = 78.065 certificates			
EUR = 78.065 x 75.36		= 5,882.98	

Independently checkable by hand: $136.4 - 64.935 = 71.465$; $71.465 + 6.6 = 78.065$; $78.065 \times 75.36 = 5,882.98$.

4. Three rules to test hardest

4.1 The zero floor applies to the direct side only

```
net = Decimal.max(0, E - F).plus(indirect)
```

Free allocation can exceed a clean producer's own emissions, where a low emitter is measured against a dirtier benchmark. The regulation surrenders certificates and never issues them, so the floor is zero. It is applied *before* indirect emissions are added, because a generous direct benchmark must not wipe out indirect emissions it was never granted against. Verified in the running tool: free allocation stayed at 64.935 tCO2e in both the direct only and the direct plus indirect runs, while the charge rose.

4.2 The CSCF is unpublished, and is never silently treated as 1.0

For 2026 the rule pack holds `value: null, status: pending`. The engine returns a distinct `cscf_pending` result whose type gives the caller nowhere to read a final value from. What is shown instead is a scenario explicitly labelled with `assumedCscf: "1"`, the last value the Commission actually set, covering 2021 to 2025. The real figure cannot be higher and may be lower.

4.3 Electricity is zero by law, not zero by arithmetic

IR (EU) 2025/2620 Art 2(2) sets electricity's free allocation to nil outright. The engine returns a separate `zero_by_fiat` status. An earlier version returned an ordinary priced result carrying a fabricated `cscf: '0'`, and the interface reported "CSCF published, CSCF 0", which was a false statement about a regulation whose

CSCF is unpublished. A zero set by law and a zero produced by calculation are different claims and no longer share a data shape.

5. Lookup discipline: fail closed

A miss raises `REGULATION_NOT_FOUND`. A tie raises `REGULATION_AMBIGUOUS`. Never a default, never a first match.

- **Route fallback.** An exact route match wins. Failing that, a row with an empty route indicator is route independent in the Annex and applies to any route. Where the Annex does enumerate routes and none matches the declared one, there is no fallback row and the lookup fails closed.
- **CN fallback.** Some goods are listed at heading level rather than at eight digits, so the most specific listed prefix wins. An equal length tie is ambiguous rather than arbitrary.
- **Precursors are collected before multiplying.** If one precursor benchmark is missing the whole SEFA becomes unavailable, rather than silently becoming a smaller deduction. A smaller deduction is a larger bill charged on our authority.

At present **183 of the 574 goods offered resolve to no figure**, and 181 of those are iron and steel.

6. Arithmetic and rounding

All arithmetic uses `decimal.js` configured at **precision 34, rounding `ROUND_HALF_EVEN`**. This is deliberately identical to the GeoCBAM ledger's configuration so the two implementations cannot disagree even at the 34th significant digit. Currency alone uses `toFixed(2, ROUND_HALF_UP)`.

Certificate rounding is not settled in law. The engine therefore reports decimal certificate equivalents rather than inventing a rounding rule.

7. Disclosed limitations

- **Column A and Equation 4 are not reachable from this tool.** See section 2.2. The code exists and is audited; the input mode is deliberately absent.
- **Article 9 is not modelled.** The deduction for a carbon price already paid in the country of origin is omitted because the implementing act remains a draft. Figures are therefore conservative, that is, they may overstate exposure.
- **Residual origin values are flagged.** Where the Commission publishes no default for a given origin, the "Other Countries and Territories" residual is used, and the output says so on the same surface as the number.
- **De minimis threshold.** Reg (EU) 2023/956 Art 2(3) as amended by Reg (EU) 2025/2083 sets 50 tonnes per importer per calendar year, covering cement, iron and steel, aluminium and fertilisers. Electricity and hydrogen are excluded from the exemption. A single customs line can only ever prove the "above threshold" case; below that the state is reported as indeterminate, never as exempt, because one line is not an annual total.

8. Source files for review

FILE	LINES	CONTENTS
cbam/sefa.ts	234	Equations 2, 4 and 6; Column A/B selection; CSCF handling
cbam/certificate-estimate.ts	352	figureFrom, the zero floor, the result union
cbam/resolve-fa.ts	164	Fail closed lookups, route and CN fallback
estimator/estimate-from-pack.ts	374	Factor selection, mark up, threshold, and the fixed scope in section 2.2

All four sit under `src/scripts/cbam-algos/`.

9. Provenance and verification

- The engine is **byte identical** to the GeoCBAM SaaS implementation. This is enforced mechanically by `scripts/cbam-sync-check.mjs`, which runs before every test and fails if any vendored file's hash moves or if upstream has advanced.
- Every rule value in the pack carries a `sourceLocator` naming the instrument and article it came from. These appear in the tool's provenance stamp on every result, including refusals.
- 50 unit tests pin the figures in section 3 and assert, per result branch, both what must appear and what must not. A refusal that quietly rendered a zero is the failure mode those tests exist to prevent.

Known gap, disclosed. Two of the four result branches, `ok` and `zero_by_fiat`, are currently unreachable with the shipped rule pack. No CSCF is published for any year that also has production routes, so every 2026 line resolves to `cscf_pending`; and the pack contains no electricity classifications at all. Both paths are correct by inspection and covered by unit tests against synthetic input, but neither is exercised by real data today.

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